Proving triangles congruent

1

a 90°

b They are both right-angled triangles with hypotenuse and one leg the same length.

c Hypotenuse-leg congruence (HL)

2 Angle-side-angle congruence (ASA)

3 Side-side-side congruence (SSS)

4

To prove: $\triangle PQR \cong \triangle RSP$

	Statements	Reasons
1.	$\overline{PS} \cong \overline{RQ}$	Given
2.	$\overline{PR} \cong \overline{RP}$	Reflexive property of congruence
3.	$\angle QRP \cong \angle SPR$	Given
4.	$\triangle PQR \cong \triangle RSP$	Side-angle-side congruence (SAS)

5

To prove: $\triangle RTQ \cong \triangle RSP$

	Statements	Reasons
1.	$\overline{PS} \cong \overline{QT}$	Given
2.	$\angle PSR \cong \angle QTR$	Given
3.	$\angle QRT \cong \angle PRS$	Vertical angles congruence theorem
4.	$\triangle RTQ \cong \triangle RSP$	Angle-angle-side congruence (AAS)

6

To prove: $\triangle$[redacted] $\cong \triangle RSP$

	Statements	Reasons
1.	$\angle PSR \cong \angle RQP$	Given
2.	$\overline{PS} \parallel \overline{QR}$	Given
3.	$\angle SPR \cong \angle QRP$	Alternate interior angles theorem
4.	$\overline{PR} \cong \overline{RP}$	Reflexive property of congruence
5.	$\triangle PQR \cong \triangle RSP$	Angle-angle-side congruence (AAS)

7

To prove: $\triangle DEH \cong \triangle FEG$

	Statements	Reasons
1.	E is the midpoint of $\overline{DF}$	Given
2.	$\overline{DH} \cong \overline{FG}$	Given
3.	$\overline{EH} \cong \overline{EG}$	Given
4.	$\overline{DE} \cong \overline{EF}$	Properties of Midpoint
5.	$\triangle DEH \cong \triangle FEG$	Side-side-side congruence theorem (SSS)

8

To prove: $\triangle AXD \cong \triangle CXB$

	Statements	Reasons
1.	$\overline{AD} \parallel \overline{CB}$	Given
2.	$\overline{AC}$ bisects $\overline{BD}$	Given
3.	$\overline{DX} \cong \overline{BX}$	Definition of Bisect
4.	$\angle AXD \cong \angle CXB$	Vertical angles congruence theorem
5.	$\angle ADX \cong \angle CBX$	Alternate interior angles theorem
6.	$\triangle AXD \cong \triangle CXB$	Angle-side-angle congruence (ASA)

9

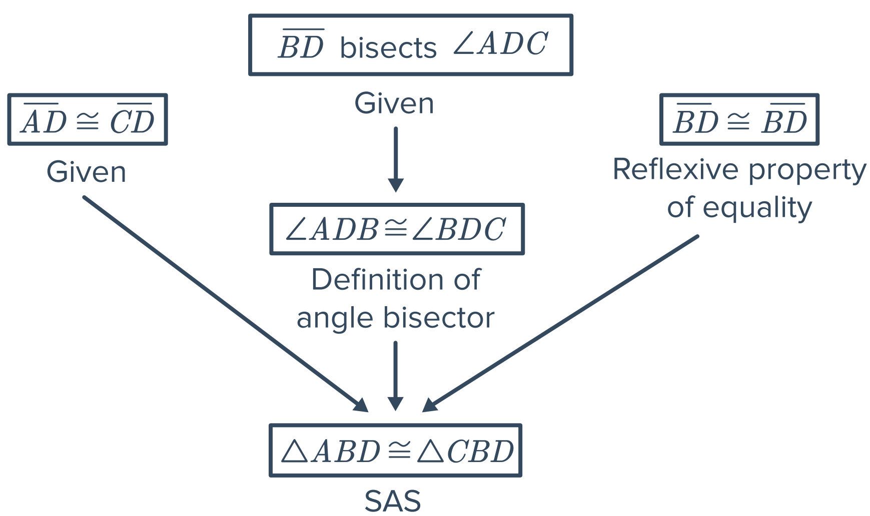
To prove: $\triangle$[redacted] $\cong \triangle MQP$

	Statements	Reasons
1.	$\overline{RQ} \cong \overline{QP}$	Given
2.	$\overline{RP} \perp \overline{MQ}$	Given
3.	$\angle MQR$ and $\angle MQP$ are right angles	Definition of perpendicular
4.	$\angle MQR \cong \angle MQP$	All right angles are congruent
5.	$\overline{MQ} \cong \overline{MQ}$	Reflexive property of congruence
6.	$\triangle MQR \cong \triangle MQP$	Side-angle-side condgruence (SAS)

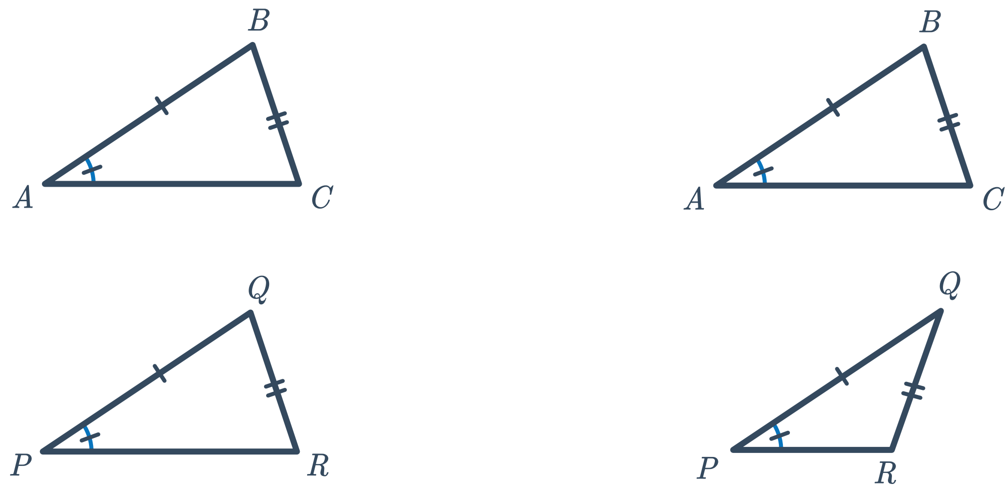
Additional questions

10 Answers may vary:
From the diagram we know that $\overline{PQ} \cong \overline{QR}$ and $\overline{PS} \cong \overline{RS}$. Then, $\overline{QS} \cong \overline{QS}$ by the reflexive property. We can now conclude that $\triangle SPQ \cong \triangle SRQ$ by Side-side-side congruence (SSS).

11 Answers may vary:



12 Consider the triangles shown in the diagram[redacted]n the given informaton there are *two possible* triangles, one of which is not congruent. So we do not have enough information to conlcude that the triangles are congruent.



13 Answers may vary:
Starting with the outer frame, Quentin should cut two congruent pieces of wood for the sides and one shorter piece of wood for the base. Then, Quentin should cut two pieces of wood that are exactly half the length of the two congruent sides as well as one piece of wood that is exactly half the width of the base. By placing these pieces at the midpoint of the sides he can guarantee that all triangles are congruent by Side-side-side congruene (SSS).

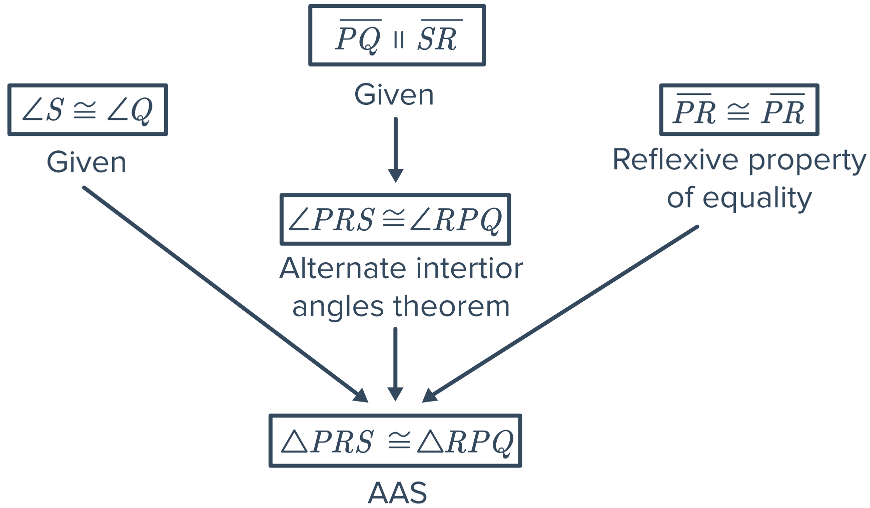
14 Answers may vary:

To prove: $\triangle ABX \cong \triangle DCX$

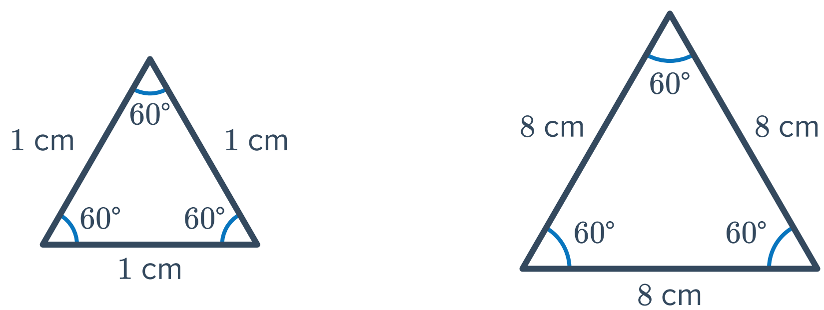
	Statements	Reasons
1.	$\overline{XB} \cong \overline{XC}$	Given
2.	$\overline{CD} \parallel \overline{AB}$	Given
3.	$\angle CDX \cong \angle BAX$	Alternate interior angles theorem
4.	$\angle AXB \cong \angle DXC$	Vertical angles theorem
5.	$\triangle ABX \cong \triangle DCX$	Angle-angle-side congruence (AAS)

An alternate proof could use the alternate interior angle theorem to show that $\angle ABX \cong \angle DCX$, thus proving the triangles congruent by Angle-angle-side congruence (AAS).

15 Answers may vary:



16 Two triangles with congruent angles don't have to have congruent size. Consider these two equilateral triangles (where all angles are 60°):



17 From the diagram we know that $\overline{MQ} \cong \overline{NP}$ by definition of congruence. We also know that $\angle N$ and $\angle Q$ are right angles so $\triangle MQP$ and $\triangle PNM$ are both right triangles by the definition of right triangles. Next, we know that $\overline{MP} \cong \overline{MP}$ by the Reflexive property. Therefore $\triangle MQP \cong \triangle PNM$ by the Hypotenuse-Leg congruence theorem.

18 Answers may vary:

To prove: $\triangle ACD \cong \triangle ABC$

	Statements	Reasons
1.	$\overline{AD} \cong \overline{AD}$	Reflexive property of equality
2.	$\triangle ABC$ is equilateral	Given
3.	$\overline{AC} \cong \overline{AB}$	Definition of equilateral
4.	$m\angle ADB = 90^\circ$	Given
5.	$m\angle ADC + m\angle ADB = 180^\circ$	Linear pair postulate
6.	$m\angle ADC + 90^\circ = 180^\circ$	Substitution
7.	$m\angle ADC = 90^\circ$	Subtraction property of equality
8.	$\triangle ACD$ and $\triangle ABD$ are right-triangles	Definition of right-triangle
9.	$\triangle PQR \cong \triangle RSP$	Hypotenuse-leg (HL) congruency theorem

19 Answers may vary:

To prove: $\triangle HJK \cong \triangle LKJ$

	Statements	Reasons
1.	$\overline{HJ} \parallel \overline{KL}$	Given
2.	$m\angle HJK = 90^\circ$	Given
3.	$\angle HJK \cong \angle LKJ$	Alternate interior angles theorem
4.	$m\angle LKJ = 90^\circ$	Congruent angles have equal measures
5.	$\triangle HJK$ and $\triangle LKJ$ are right-triangles	Definition of right-triangle
6.	$\overline{HK} \cong \overline{JL}$	Given
7.	$\overline{JK} \cong \overline{KJ}$	Reflexive property of equality
8.	$\triangle ABX \cong \triangle DCX$	Hypotenuse-leg (HL) congruency theorem

20 Answers may vary:
We are given that $\overline{AE}$ bisects $\overline{BD}$ so we know that $\overline{BC} \cong \overline{DC}$ by the definition of a segment bisector. We can also conclude from the diagram that $\angle ACB$ and $\angle ECD$ are vertical angles, which tells us that $\angle ACB \cong \angle ECD$, and so $m\angle ACB = m\angle DCE = 90^\circ$ by definition of congruence. So we know that both triangles are right-triangles. Finally, we are given that $\overline{AB} \cong \overline{DE}$. Thus $\triangle ABC \cong \triangle EDC$ by the Hypotenuse-leg (HL) congruency theorem.

21 The four corner angles of a rectangle are each right-angles, so we know that both triangles are right-triangles. The hypotenuse of the two triangles are the same length by the reflexive property of equality. We also know that opposite sides of a rectangle have equal measures, so we can choose either pair of opposite sides to find a pair of congruent legs between the triangles. Putting this together, we can conclude that the triangles are congruent by HL.